A Multivariate Statistical Framework for FX Market Microstructure: Behavioral Segmentation of Corporate Clients

Corporate FX Sales & Solutions · Markets · Citibank Colombia S.A.

Jose Miguel León Puentes
Departamento de Estadística
Facultad de Ciencias
Universidad Nacional de Colombia

Oscar Orlando Melo Martínez
Departamento de Estadística
Facultad de Ciencias
Universidad Nacional de Colombia

Abstract

This research develops a multivariate statistical framework for the behavioral segmentation of corporate Foreign Exchange (FX) clients. To address data confidentiality, a rigorous three-level hierarchical Data Generation Process (DGP) was designed to replicate institutional market microstructures, including heavy-tailed distributions and zero-inflation in spreads. The analytical pipeline utilized Principal Component Analysis (PCA) to reduce 14 independent dimensions into four components (explaining 69.46% of variance) and applied unsupervised clustering. Validated with an Adjusted Rand Index of 0.9722, the model identified two distinct segments: Corporate (19.6%) and Transactional (80.4%). While both groups share similar digital channel preferences, they differ drastically in scale, economic value, and operational consistency. These findings provide a quantitative foundation for transitioning commercial management from informal intuition to evidence-based strategy.

Introduction

The corporate Foreign Exchange (FX) market is data-rich but often suffers from an "analytical gap". While every transaction is recorded, commercial management frequently relies on intuition or informal volume-based classifications rather than structured statistical knowledge. In emerging markets, local teams often lack access to advanced quantitative methodologies due to the strategic confidentiality of transactional data. This research bridges that gap by building a rigorous quantitative portrait of a client portfolio using unsupervised multivariate analysis

Objectives

- **General:** Develop a multivariate analytical pipeline for the behavioral segmentation of corporate FX clients using a Data Generation Process (DGP) that reproduces market microstructures
- **Specific:**
  - Design a hierarchical conditional DGP that simulates heavy-tailed distributions, economies of scale, and behavioral heterogeneity.
  - Apply Principal Component Analysis (PCA) to reduce dimensionality while maintaining geometric quality.
  - Statistically characterize segments to derive actionable commercial strategies

Methodology

- **Data Generation (DGP)**: Because real FX data is confidential, a three-level hierarchical simulation was built, creating 500 clients and 34,936 transactions with realistic stylized facts (zero-inflation in spreads, heavy tails, and Pareto effects).
- **Feature Engineering**: A vector of 14 genuinely independent dimensions was created, neutralizing algebraic dependencies found in accounting data.
- **Analytical Pipeline:**
  - **Dimensionality Reduction:** PCA identified four components explaining 69.46% of the variance (KMO = 0.716).
  - **Clustering:** A comparison of four algorithms (K-Means, Ward, Hybrid, and PAM) was performed.
  - **Outlier Diagnosis:** Used DBSCAN to identify and exclude "noise" observations (clients with minimal activity).
  - **Validation:** External validation against the DGP's known structure yielded an Adjusted Rand Index (ARI) of 0.9722, confirming near-perfect recovery of client archetypes

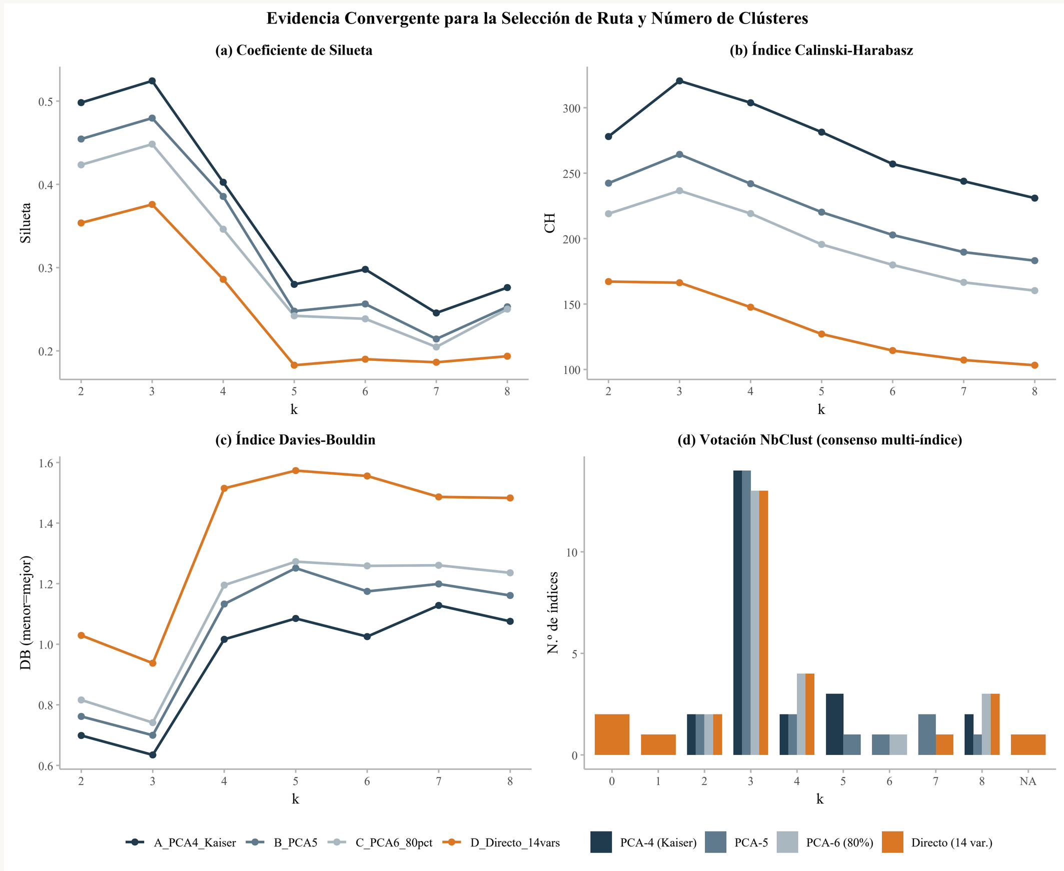
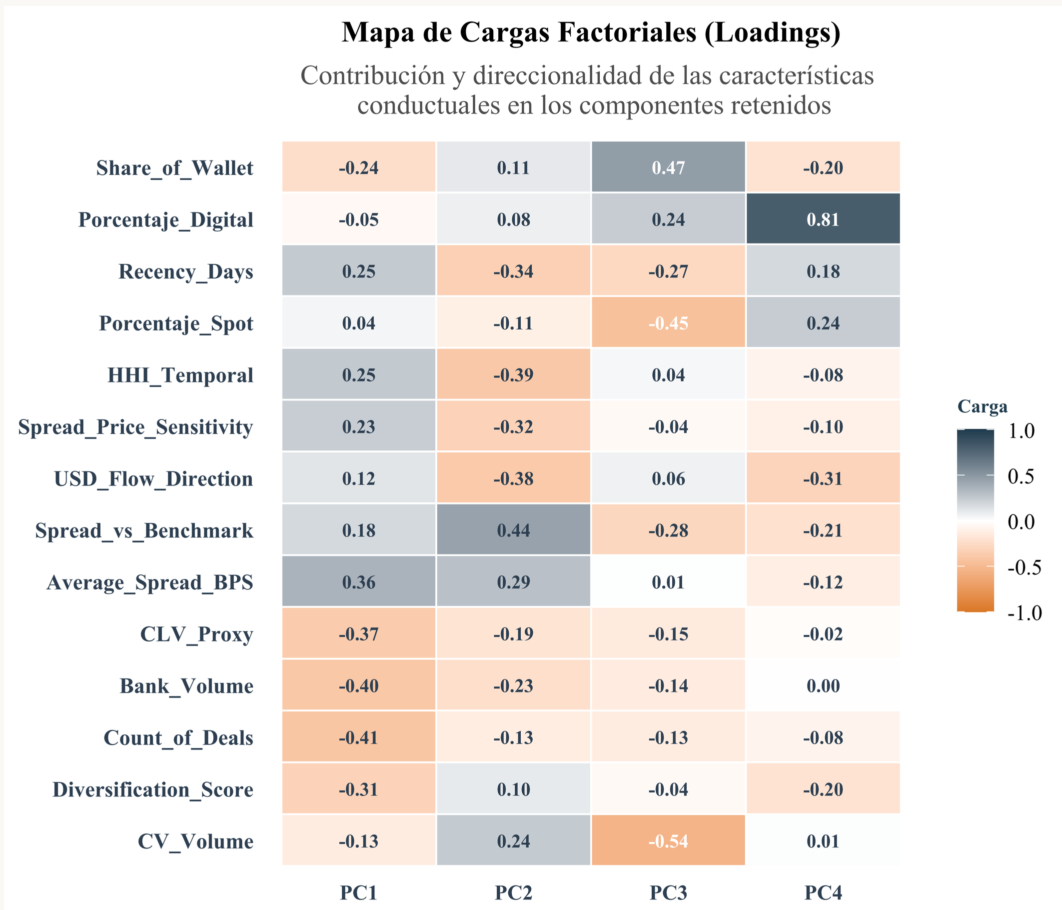


Fig. 1. Panel de evidencia de clustering: dendrograma jerárquico, curva de silhouette y proyección UMAP coloreada por segmento.



	PC1	PC2	PC3	PC4
Share_of_Wallet	-0.24	0.11	0.47	-0.20
Porcentaje_Digital	-0.05	0.08	0.24	0.81
Recency_Days	0.25	-0.34	-0.27	0.18
Porcentaje_Spot	0.04	-0.11	-0.45	0.24
HHI_Temporal	0.25	-0.39	0.04	-0.08
Spread_Price_Sensitivity	0.23	-0.32	-0.04	-0.10
USD_Flow_Direction	0.12	-0.38	0.06	-0.31
Spread_vs_Benchmark	0.18	0.44	-0.28	-0.21
Average_Spread_BPS	0.36	0.29	0.01	-0.12
CLV_Proxy	-0.37	-0.19	-0.15	-0.02
Bank_Volume	-0.40	-0.23	-0.14	0.00
Count_of_Deals	-0.41	-0.13	-0.13	-0.08
Diversification_Score	-0.31	0.10	-0.04	-0.20
CV_Volume	-0.13	0.24	-0.54	0.01

Fig. 2. Heatmap de cargas PCA. Las primeras cuatro componentes explican el 71.3% de la varianza total. Los colores indican la dirección e intensidad de cada carga.

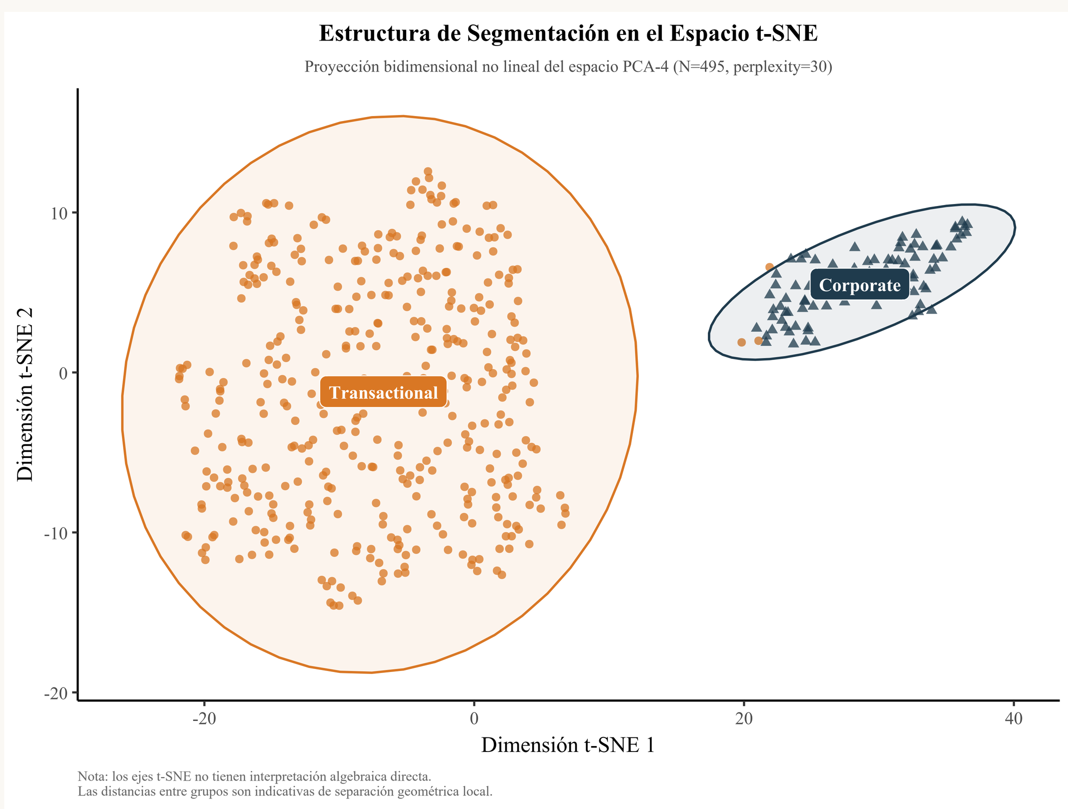


Fig. 3. Perfiles medios de los cinco segmentos identificados. Cada eje representa una dimensión de comportamiento normalizada.

Results

- Segment Identification: Two robust segments were identified: Corporate (n = 97, 19.6%) and Transactional (n = 398, 80.4%).
- Scale vs. Nature: Segments differ drastically in scale (Corporate median volume: $363.9M vs. Transactional: $4.7M) and consistency. However, they are statistically indistinguishable in their choice of channel (80% digital for both), instrument (95% spot), and currency pairs.
- Microeconomics of Spread: Validated economies of scale (R^2 = 92.23%). For every 1% increase in volume, revenue increases by only 0.65%, confirming that spreads compress as volume grows.
- Portfolio Concentration: The market exhibits extreme concentration, with a Gini coefficient of 0.81 for volume.

Key Finding 5 segmentos comportamentales robustos

Silhouette = 0.61 · **Varianza explicada** = 71.3%

El segmento S2 concentra el 63% del volumen total con solo el 12% de los clientes

Discussion

- **Price Strategy:** Corporate clients have high price sensitivity and operate near the "benchmark" spread, while Transactional clients pay a higher median spread (161.5 bps vs. 47.0 bps), suggesting more room for value capture in the smaller segment.
- **Digitalization:** Since digital adoption is high across both segments (80%), technology should focus on real-time pricing intelligence for Corporate clients and operational simplicity for Transactional ones.
- **Relationship Management:** Suggests a split in sales force: Relationship Managers for Corporate (complex, high-frequency) and Flow Traders for Transactional (speed and efficiency)

Conclusions

- The study successfully transitioned informal commercial intuition into evidence-based statistical segmentation.
- The behavioral heterogeneity of the portfolio is driven primarily by intensity and regularity rather than the type of products used.
- The methodology provides a robust framework that can be directly applied to real-world institutional banking data

Trabajo de Grado
Modalidad: Pasantía
Pregrado en Estadística

Repositorio del proyecto

Contacto
jmlionc.github.io
joleonp@unal.edu.co